Telescope Calculations You Will Actually Use

What you will be able to do

Given a telescope's **aperture** D and **focal length** F and an eyepiece of focal length f, calculate:

1. the focal ratio;
2. the magnification, and the eyepiece needed for a target magnification;
3. the maximum and minimum useful magnification;
4. the exit pupil;
5. the true field of view;
6. the light-gathering power;
7. the resolving power;
8. the limiting magnitude;
9. the image scale for cameras (bonus).

Our two example telescopes

Telescope A: 114 mm Newtonian, $F = 900$ mm (a common first telescope). **Telescope B:** 200 mm (8-inch) Dobsonian, $F = 1200$ mm (a club telescope).
Eyepieces: 25 mm and 10 mm Plössl (apparent field 50°–52°), and a 2× Barlow lens.

1. Focal ratio

$$\boxed{N = \frac{F}{D}} \qquad A: \ \frac{900}{114} = \mathbf{f/7.9} \qquad B: \ \frac{1200}{200} = \mathbf{f/6}$$

Fast (f/4–f/6): wider fields, brighter images for photography. **Slow** (f/10–f/15): narrow fields, easy high power for planets.

2. Magnification

$$\boxed{M = \frac{F}{f}} \qquad \text{with a Barlow:} \qquad M = \frac{F \times (\text{Barlow factor})}{f}$$

Telescope	25 mm	10 mm	10 mm + 2× Barlow
A ($F = 900$)	36×	90×	180×
B ($F = 1200$)	48×	120×	240×

Choosing an eyepiece: $f = F/M$. For $\sim$ 150× on Telescope B: $f = 1200/150 = \mathbf{8}$ mm. The *eyepiece* changes the magnification; aperture and focal length belong to the *telescope*.

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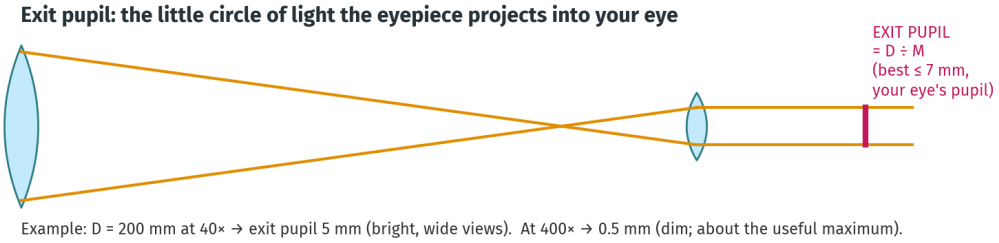
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3. How much magnification is useful?

$$M_{\max} \approx 2 \times D \text{ (mm)} \qquad M_{\min} \approx \frac{D}{7 \text{ mm}}$$

- $M_{\max}$: A ≈ 228×, B ≈ 400×. Beyond this the image grows but gets no sharper ("empty magnification"). In practice the atmosphere ("seeing") limits most nights to **150–250**×; on the hazy Indian plains often 100–180×; more is possible in Ladakh.
- $M_{\min}$: A ≈ 16×, B ≈ 29×. Below this the beam leaving the eyepiece is wider than your dark-adapted pupil (~7 mm) and light is wasted (older observers' pupils open to only 5–6 mm).

4. Exit pupil



The exit pupil is the little circle of light the eyepiece sends into your eye.

$$\text{exit pupil} = \frac{D}{M} = \frac{f}{N} \qquad A + 25 \text{ mm}: \ \frac{114}{36} = 3.2 \text{ mm} \left(= \frac{25}{7.9} \checkmark \right)$$

Exit pupil	Image	Good for
5–7 mm	brightest, widest	galaxies, nebulae, star fields
2–4 mm	balanced	clusters, general viewing
0.5–1.5 mm	dim, magnified	planets, the Moon, double stars
< 0.5 mm	too dim	avoid

5. True field of view: will it fit?

$$\text{TFOV} \approx \frac{\text{AFOV}}{M}$$

(AFOV = apparent field of the eyepiece: ~ 52° Plössl, 60°–82° wide-field)

	M	TFOV (52° eyepiece)
A + 25 mm	36×	52/36 = 1.44°
A + 10 mm	90×	0.58°
B + 25 mm	48×	1.08°
B + 10 mm	120×	0.43°

The Moon (0.5°) fits in all of these except B + 10 mm. The Pleiades (~ 2°) do not fit even in A + 25 mm — use binoculars! The Andromeda Galaxy is ~ 3° long.

6. Light-gathering power

Light collected $\propto$ area $\propto D^2$:

$$\boxed{\text{LGP vs eye} = \left(\frac{D}{7\text{ mm}}\right)^2} \qquad A:\ \left(\tfrac{114}{7}\right)^2 \approx 265\times \qquad B:\ \left(\tfrac{200}{7}\right)^2 \approx 816\times \qquad \frac{B}{A} = \left(\tfrac{200}{114}\right)^2 \approx 3.1$$

7. Resolving power

Light is a wave, so even a perfect telescope blurs a star into a tiny disc; two stars closer than a certain angle merge.

$$\boxed{\text{Dawes limit: } R \approx \frac{116}{D\ (\text{mm})}\ \text{arcsec}} \qquad \left(\text{Rayleigh: } R = 1.22\,\frac{\lambda}{D} \approx \frac{138}{D\ (\text{mm})}\ \text{arcsec at } 550\text{ nm}\right)$$

A: $116/114 \approx 1.0''$ B: $116/200 \approx 0.58''$ eye (7 mm): $\approx 17''$ in theory, $\sim 60''$ in practice.

Double stars: the "Double Double" (ε Lyrae, near Vega) has pairs $2.3''$ and $2.4''$ apart — both telescopes split them at $\sim$ 150× on a steady night. Mizar's companion is $14''$ away — easy at 36×. The atmosphere usually blurs images to $1''$–$3''$.

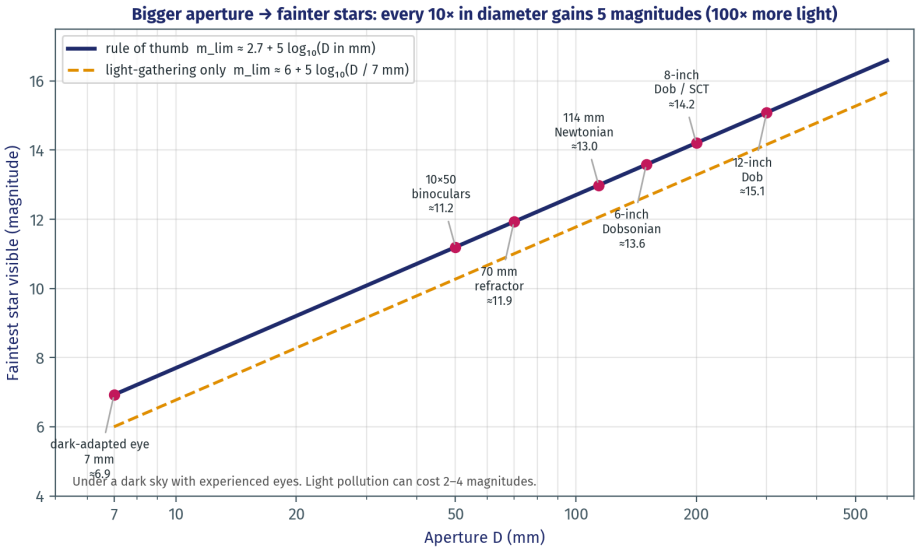
8. Limiting magnitude: the faintest star you can see

The telescope collects $(D/7)^2$ times more light than your eye. Using the magnitude formula of Lecture 3.1, that is a gain of

$$\Delta m = 2.5\log_{10}\left(\frac{D}{7}\right)^2 = 5\log_{10}\frac{D}{7} \quad\Longrightarrow\quad m_{\text{lim}} \approx 6 + 5\log_{10}\frac{D}{7\text{ mm}}$$

Observers commonly use the slightly more generous rule of thumb (higher magnification darkens the sky background):

$$\boxed{m_{\text{lim}} \approx 2.7 + 5\log_{10} D\ (\text{mm})} \qquad A:\ 2.7 + 5\times 2.057 = \mathbf{13.0} \qquad B:\ 2.7 + 5\times 2.301 = \mathbf{14.2}$$



Every 10× in diameter gains 5 magnitudes (100× more light). Under city skies subtract 1–3 magnitudes.

Worked example

Can Telescope B show Pluto ($m = 14.4$)? $m_{\text{lim}} = 14.2 < 14.4$, so **no** (only perhaps under a perfect sky with an expert eye). You need $D = 10^{(14.4-2.7)/5} \approx 220\,\text{mm}$ or more — or a camera.

9. Image scale for cameras (bonus)

With a camera at the telescope's focus (no eyepiece):

$$\text{image size} = F \times \theta\ (\text{radians}), \qquad \text{plate scale} = \frac{206{,}265}{F\ (\text{mm})}\ \frac{\text{arcsec}}{\text{mm}},$$

$$\text{pixel scale} = \frac{206.265 \times p\ (\mu\text{m})}{F\ (\text{mm})}\ \frac{\text{arcsec}}{\text{pixel}}$$

The Moon ($0.52^\circ = 0.00908\,\text{rad}$) on Telescope A: $900 \times 0.00908 \approx 8.2\,\text{mm}$ — fits a typical $22 \times 15\,\text{mm}$ sensor. With $3.76\,\mu\text{m}$ pixels: $206.265 \times 3.76/900 \approx 0.86''$ per pixel.

10. One-page summary

Quantity	Formula	A (114/900)	B (200/1200)
Focal ratio	F/D	f/7.9	f/6
Magnification	F/f	36× (25 mm), 90× (10 mm)	48×, 120×
Max / min useful	$2D$ / $D/7$	~ 228× / ~ 16×	~ 400× / ~ 29×
Exit pupil (25 mm)	D/M	3.2 mm	4.2 mm
True field (25 mm)	AFOV$/M$	1.44°	1.08°
Light vs eye	$(D/7)^2$	265×	816×
Dawes limit	$116/D$	1.0″	0.58″
Limiting magnitude	$2.7 + 5\log_{10} D$	13.0	14.2

Using the numbers

- **"The whole Moon in Telescope B."** Need TFOV $> 0.5^\circ \Rightarrow M < 52/0.5 \approx 100\times$ — use the 25 mm (48×).
- **"The Orion Nebula."** Large and faint: big exit pupil, low power — the 25 mm.
- **"Saturn's rings in Telescope A?"** The rings span ~ 40″; visible from ~ 25×, clear at 90× — yes!
- **"My 60 mm telescope says up to** 675×**."** $M_{\text{max}} \approx 120\times$ — the rest is empty magnification.

Key points to remember

- $N = F/D$; $M = F/f$; $M_{\text{max}} \approx 2D$; $M_{\text{min}} \approx D/7$; exit pupil $= D/M$; TFOV $=$ AFOV$/M$.
- LGP $\propto D^2$; Dawes $= 116/D$ arcsec; $m_{\text{lim}} \approx 2.7 + 5\log_{10} D$.
- Aperture decides light grasp, resolution and limiting magnitude; the eyepiece only changes magnification and field.
- The atmosphere limits most nights to ~ 150–250× and ~ 1″–2″.

Check yourself

1. An 80 mm f/5 refractor: focal length? Magnification with a 20 mm eyepiece?
2. Its maximum useful magnification and limiting magnitude?
3. Which collects more light: one 150 mm telescope, or two 100 mm telescopes together?
4. Can a 50 mm telescope split a double star 2″ apart?

Answers: 1. $F = 400$ mm; $M = 20\times$. 2. 160×; $2.7 + 5\log_{10} 80 \approx 12.2$. 3. $150^2 = 22{,}500 > 2 \times 100^2 = 20{,}000$: the 150 mm. 4. Dawes $= 116/50 = 2.3'' > 2''$: not quite.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §6.1 "Telescopes" — https://openstax.org/books/astronomy-2e/pages/6-1-telescopes
- NCERT *Physics* Class XII Part II, Ch. 9 "Ray Optics and Optical Instruments" — magnifying power of telescopes.